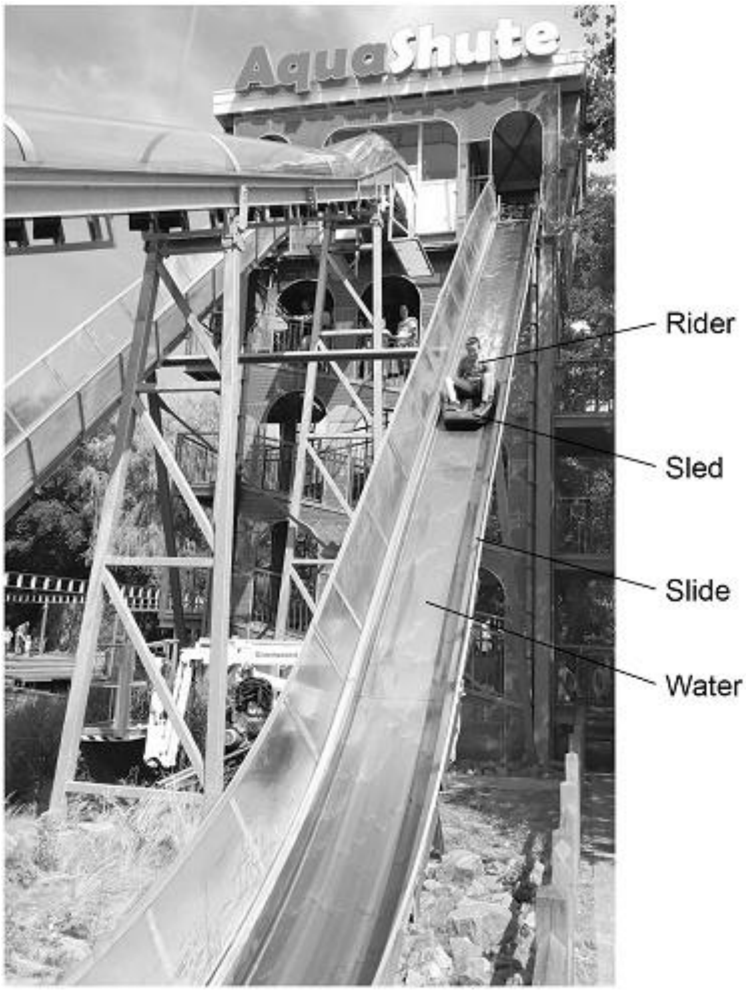
Q1.

The photograph below shows a theme park ride called AquaShute.



(a) Riders of the AquaShute sit on a sled and move down a slide.

There is a layer of water between the sled and the slide.

How does the layer of water affect the friction between the sled and the slide?

Tick (✓) **one** box.

The friction is decreased.	☐
The friction is increased.	☐
The friction is not affected.	☐

(1)

(b) The mass of one rider is 62.5 kg.

The height of the slide is 16.0 m.

gravitational field strength = 9.8 N/kg

Calculate the gravitational potential energy of the rider at the top of the slide.

Use the equation:

gravitational potential energy = mass × gravitational field strength × height

___

___

___

___

Gravitational potential energy = ________________ J

(2)

(c) At the bottom of the slide the speed of the rider is 12 m/s.

The mass of the rider is 62.5 kg.

Calculate the kinetic energy of the rider at the bottom of the slide.

Use the equation:

kinetic energy = 0.5 × mass × $(\text{speed})^2$

___

___

___

___

Kinetic energy = ________________ J

(2)

(d) When a rider reaches the bottom of the slide, the sled decelerates and stops.

Give **two** factors that will affect how far the sled will move before it stops.

1 ___

___

2 ___

___

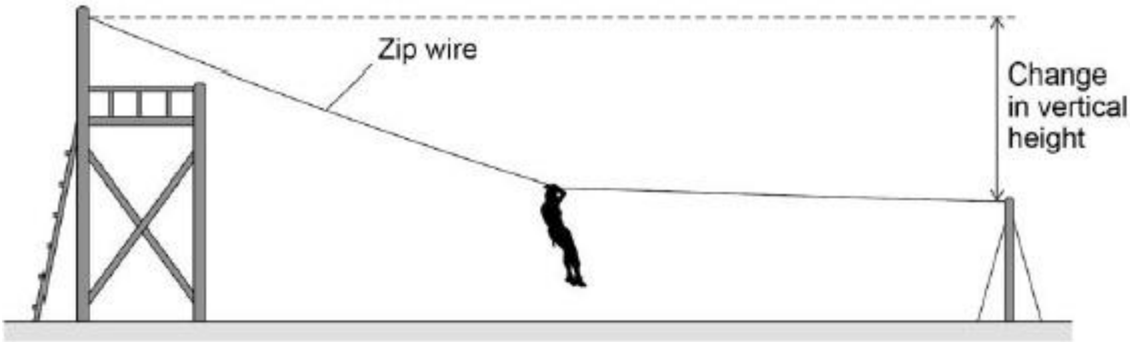
(2)

(Total 7 marks)

Q2.

Figure 1 shows a person sliding down a zip wire.

Figure 1



(a) As the person slides down the zip wire, the change in the gravitational potential energy of the person is 1.47 kJ

The mass of the person is 60 kg

gravitational field strength = 9.8 N/kg

Calculate the change in vertical height of the person.

____

____

____

____

____

Change in vertical height = ______________________________ m

(3)

(b) As the person moves down the zip wire her increase in kinetic energy is less than her decrease in gravitational potential energy.

Explain why.

____

____

____

____

____

(2)

(c) Different people have different speeds at the end of the zip wire.

Explain why.

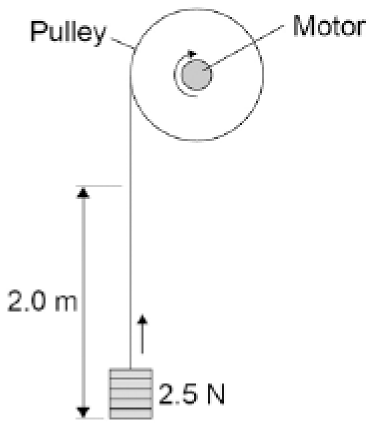
(2)

(Total 7 marks)

Q3.

A student investigated the efficiency of a motor using the equipment in **Figure 1**.

Figure 1



He used the motor to lift a weight of 2.5 N a height of 2.0 m.

He measured the speed at which the weight was lifted and calculated the efficiency of the energy transfer.

He repeated the experiment to gain two sets of data.

(a) Give **one** variable that the student controlled in his investigation.

___

(1)

(b) Give **two** reasons for taking repeat readings in an investigation.

1. ___

___

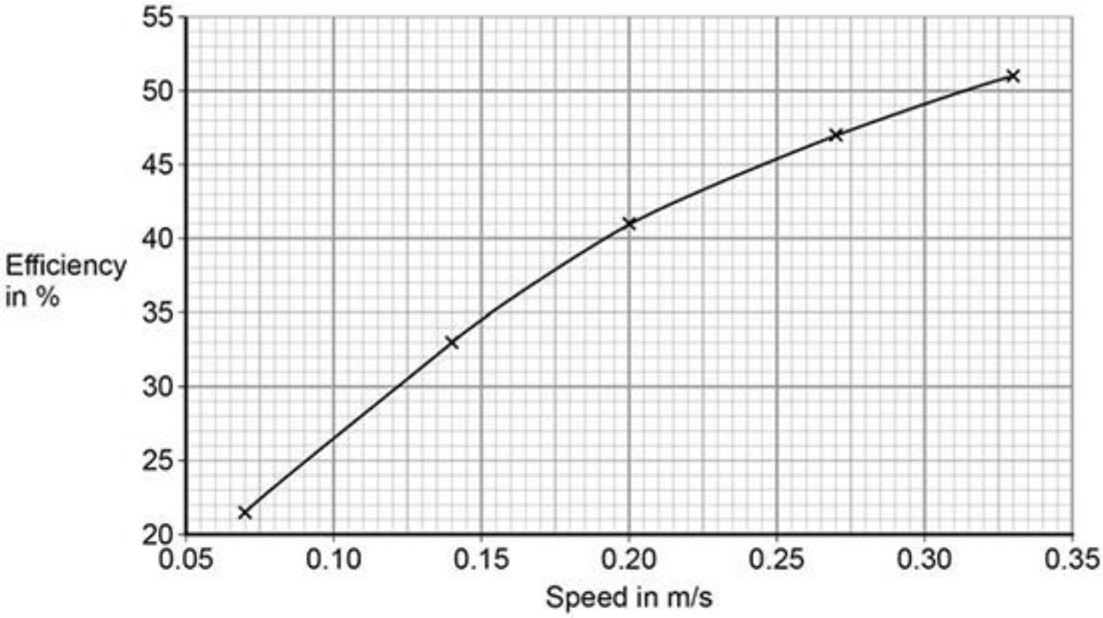
2. ___

___

(2)

(c) **Figure 2** shows a graph of the student's results.

Figure 2



Give **two** conclusions that could be made from the data in **Figure 2**.

____

____

____

____

(2)

(d) Give the main way that the motor is likely to waste energy.

____

____

(1)

(e) When the total power input to the motor was 5 W the motor could not lift the 2.5 N weight.

State the efficiency of the motor.

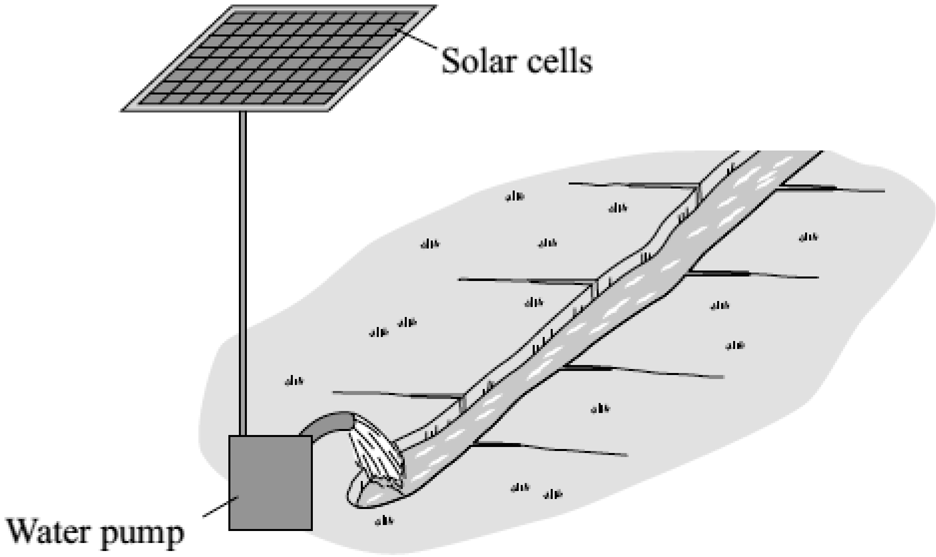
Efficiency = ________________________ %

(1)

(Total 7 marks)

Q4.

The farmers in a village in India use solar powered water pumps to irrigate the fields.



On average, a one square metre panel of solar cells receives 5 kWh of energy from the Sun each day.
The solar cells have an efficiency of 0.15

(a) (i) Calculate the electrical energy available from a one square metre panel of solar cells.

Show clearly how you work out your answer.

____

____

Electrical energy = ____________________________ kWh

(2)

(ii) On average, each solar water pump uses 1.5 kWh of energy each day.

Calculate the area of solar cells required by one solar water pump.

Area = _______________________________ square metres

(1)

(b) Give **one** reason why the area of solar cells needed will probably be greater than the answer to part (a)(ii).

____

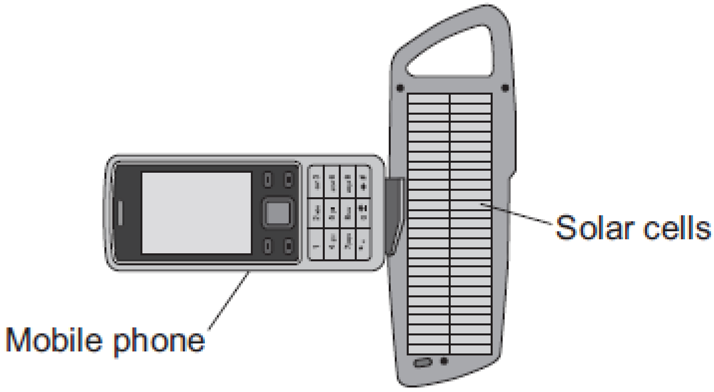
____

(1)

(Total 4 marks)

Q5.

(a) The diagram shows a solar powered device being used to recharge a mobile phone.



On average, the solar cells produce 0.6 joules of electrical energy each second.
The solar cells have an efficiency of 0.15.

(i) Calculate the average energy input each second to the device.

Show clearly how you work out your answer.

____

____

Average energy input each second = ____________________ J/s

(2)

(ii) Draw a labelled Sankey diagram for the solar cells.
The diagram does **not** need to be drawn to scale.

(1)

(b) Scientists have developed a new type of solar cell with an efficiency of over 40 %.
The efficiency of the solar cell was confirmed independently by other scientists.

Suggest why it was important to confirm the efficiency independently.

____

____

(1)

(c) The electricity used in homes in the UK is normally generated in a fossil fuel power station.

Outline some of the advantages of using solar cells to generate this electricity.

____

(2)
(Total 6 marks)

Q6.

All European Union countries are expected to generate 20% of their electricity using renewable energy sources by 2020.

The estimated cost of generating electricity in the year 2020 using different energy sources is shown in **Table 1**.

Table 1

Energy source	Estimated cost (in the year 2020) in pence per kWh
Nuclear	7.8
Solar	25.3
Tidal	18.8
Wind	10.0

France generated 542 billion kWh of electricity using nuclear power stations in 2011. France used 478 billion kWh of electricity and sold the rest of the electricity to other countries in 2011.

(a) France may continue generating large amounts of electricity using nuclear power stations instead of using renewable energy resources.

Suggest **two** reasons why.

1. ____________________

2. ____________________

(2)

(b) Give **two** disadvantages of generating electricity using nuclear power stations.

1. ____________________

2. ____________________

(2)

(c) A panel of solar cells has an efficiency of 0.15.

The total power input to the panel of solar cells is 3.2 kW.

Calculate the useful power output of this panel of solar cells in kW.

____

____

____

____

Useful power output = _______________ kW

(2)

(d) **Table 2** shows the manufacturing cost and efficiency of different types of panels of solar cells.

Table 2

Type of Solar Panel	**Cost to manufacture a 1 m^2 solar panel in £**	**Efficiency in %**
A	40.00	20
B	22.50	15
C	5.00	10

Some scientists think that having a low manufacturing cost is more important than improving the efficiency of solar cells.

Use information from **Table 2** to suggest why.

____

____

____

____

____

____

(2)

(Total 8 marks)

Q7.

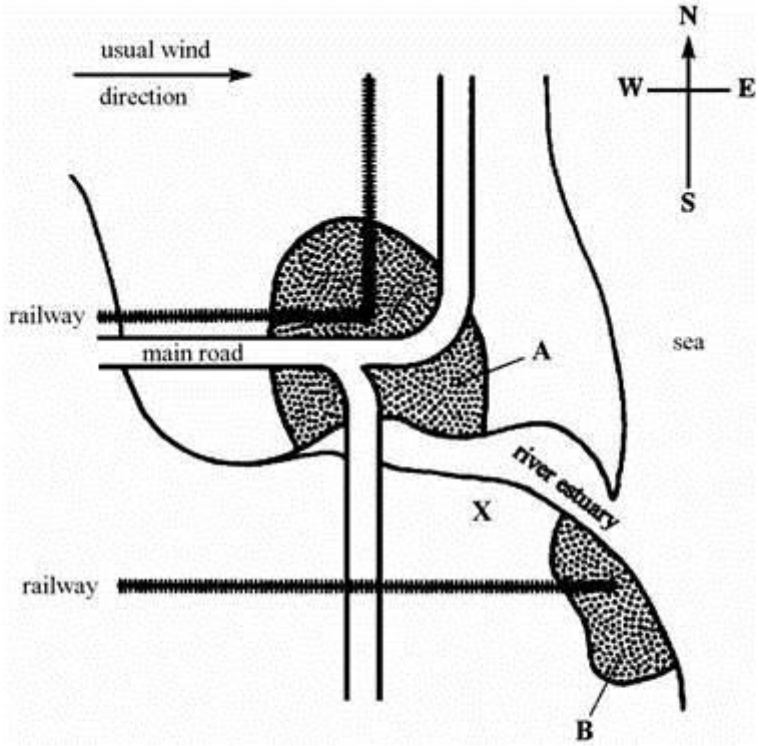
The map below shows the position of two towns, **A** and **B**, on the banks of a large river estuary.

A is an important fishing and ferry port.

The wind usually blows from the west. The major roads and railways are shown.

A power station is to be built in area X to generate electricity for the region.

The choice is between a nuclear power station and a coal fired power station.



(a) State the advantages and disadvantages of the two methods of generating electrical energy.

____

____

____

____

____

____

____

____

____

____

____

____

(6)

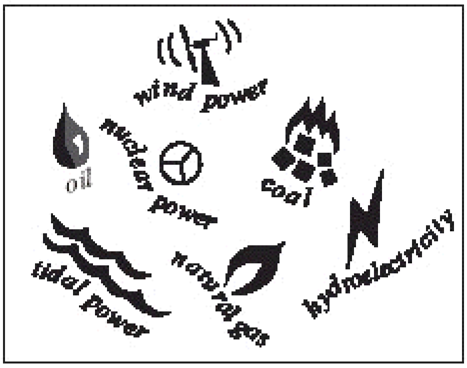
(b) Which method would you choose for this site?

Explain the reason for your choice.

(3)
(Total 9 marks)

Q8.

Different energy sources are shown in the box.



An 'Eco-home' is one which is friendly to the environment.
Imagine you are designing an 'Eco-home' which can use any of the energy sources above to generate electricity

(a) Choose **one** non-renewable energy source from the box above that could provide the electricity supply to your 'Eco-home', but which would be **unsuitable**.

Write the energy source in the table and explain, as fully as you can, why it is **unsuitable** for an 'Eco-home'.

Non-renewable energy source	Unsuitable for an 'Eco-home' because

(2)

(b) Choose **two** suitable renewable energy sources from the box opposite that could provide an electricity supply to your 'Eco-home'.

Write the two energy sources in the table and describe, in as much detail as you can, the undesirable environmental effects of using these.

Renewable energy source	**Undesirable environmental effects**
1	
2	

(4)

(Total 6 marks)

Mark schemes

Q1.

(a) the friction is decreased

1

(b) $E_p = 62.5 \times 9.8 \times 16.0$

1

$E_p = 9800$ (J)

1

(c) $E_k = 0.5 \times 62.5 \times 12^2$

1

$E_k = 4500$ (J)

1

(d) Any **two** from:

- speed (at bottom of slide)
- friction (between sled and ground)
  allow mass/weight of rider (and sled)
  allow surface type
- air resistance

2

[7]

Q2.

(a)

*an answer of 2.5 (m) scores **3** marks*

$1470 = 60 \times 9.8 \times h$

this mark may be awarded if E_p is incorrectly / not converted

1

$h = \dfrac{1470}{60 \times 9.8}$

or

$h = \dfrac{1470}{588}$

this mark may be awarded if E_p is incorrectly / not converted

1

$h = 2.5$ (m)

this answer only

1

(b) (work done against) air resistance
or
(work done against) friction (between zip line and pulley)

1

causes thermal energy to be transferred to surroundings

ignore sound energy

1

(c) different people have different surface areas

allow streamlining

allow body position

body size is insufficient

1

so would be affected by air resistance differently

or

initial speed may not be zero (1)

which would add to the total energy (of the system) (1)

allow people have different masses / weights (1)

so people have different terminal velocities (1)

reference to mass changing the kinetic energy or gravitational potential energy negates both these marks

1

[7]

Q3.

(a) weight (lifted)

or

height (lifted)

1

(b) any **two** from:

- calculate a mean
- spot anomalies
- reduce the effect of random errors

2

(c) as speed increases, the efficiency increases

1

(but) graph tends towards a constant value

or

appears to reach a limit

accept efficiency cannot be greater than 100%

1

(d) heating the surroundings

1

(e) 0 (%)

1

[7]

Q4.

(a) (i) 0.75

*allow **1** mark for correct transformation and substitution*

ie 0.15 = 5

2

(ii) 2

accept 1.5 ÷ their (a)(i) correctly calculated

1

(b) any **one** from:

- seasonal <u>changes</u>

  accept specific <u>changes</u> in conditions

  eg shorter hours of daylight in winter

- cloud cover

  accept idea of <u>change</u>

  must be stated or unambiguously implied

  eg demand for water will not (always) match supply of solar energy

  *do **not** accept figures are average on its own*

  *do **not** accept solar panels are in the shade*

1

[4]

Q5.

(a) (i) 4

*allow **1** mark for correct transformation and substitution*

ie $\frac{0.6}{0.15}$

substitution only scores if no subsequent steps are shown

2

(ii) diagram showing two output arrows with one arrow wider than the other with the narrower arrow labelled electrical / electricity / useful

1

(b) any **one** from:

- to check reliability / validity / accuracy
- to avoid bias

1

(c) any **two** from:

- produce no / less (air) pollution

  accept named pollutant

accept produces no waste (gases)

- energy is free
  accept it is a free resource
  *do **not** accept it is free*
- (energy) is renewable
- conserves fossil fuel stocks
- can be used in remote areas
- do not need to connect to the National Grid

2

[6]

Q6.

(a) any **two** from:

- cost per kWh is lower (than all other energy resources)
  allow it is cheaper
  ignore fuel cost
  ignore energy released per kg of nuclear fuel
- infrastructure for nuclear power already exists
  accept cost of setting up renewable energy resources is high
  accept many renewable power stations would be needed to replace one nuclear power station
  accept (France in 2011 already had a) surplus of nuclear energy, so less need to develop more renewable capacity for increased demand in the future
  accept France benefits economically from selling electricity
- more reliable (than renewable energy resources)
  accept (nuclear) fuel is readily available
  ignore destruction of habitats for renewables

2

(b) any **two** from:

- non-renewable
  allow nuclear fuel is running out
- high decommissioning costs
  accept high commissioning costs
- produces radioactive / nuclear waste
  allow waste has a long half-life
- long start-up time
- nuclear accidents have widespread implications
  allow for nuclear accident a named nuclear accident

eg Fukushima, Chernobyl
ignore visual pollution

2

(c) 0.48 (kW)

*allow **1** mark for correct substitution*
ie 0.15 = P / 3.2
*an answer of 480 W gains **2** marks*
*an answer of 48 or 480 scores **1** mark*

2

(d) the higher the efficiency, the higher the cost (per m^2 to manufacture)

accept a specific numerical example

1

more electricity could be generated for the same (manufacturing) cost using lower efficiency solar panels
or
(reducing the cost) allows more solar panels to be bought

accept a specific numerical example

1

[8]

Q7.

(a) m*ust give one advantage and one disadvantage of each to get 4 marks and 2 further scoring points*

Advantages and disadvantages relevant to:
(1) health risk
(5) cost
(6) environmental factors
(7) transport/ storage
e.g. common coal / nuclear – high cost of building both

<u>anti-nuclear</u> examples
nuclear fuel transported on roads/rail in region
possible effects on public health in surrounding area
high cost of de-commissioning
long life very active waste materials produced
how waste materials stored safely for a long time

<u>anti-coal</u> examples
unsightly
pollution
supplies of fuel limited
acid rain
non-renewable

<u>pro-nuclear</u> examples
fuel cheap
no foreseeable fuel shortage

<u>pro-coal</u> examples
safe
reliable
large coal reserves

disposal of solid waste is easier

to max 6

6

(b) choice 0 marks

any three valid reasons each with explanation, which may or may not be comparisons with other fuel

But

at least two of which must be relevant to this site

3

[9]

Q8.

(a) **(oil / natural gas / coal)**

no marks are given for choosing the correct non-renewable energy source

burning releases carbon dioxide (1) greenhouse effect (1)

OR

allow 2 effects for 2 marks

burning (releases sulphur dioxide (1) acid rain (1)

OR

(nuclear power)

no marks given for choosing the correct non-renewable energy source

accidents can release very dangerous radioactive material (1)

produces waste that stays dangerously radioactive for thousands of years **or** radioactive waste has to be stored safely for thousands of years (1)

accept the cost of installation and decommissioning is high

2

(b) any four from:

(wind power)

no marks are given for choosing the correct non-renewable energy source

- considered unsightly / visual pollution (1) very large areas of land (1)
- noisy for people living nearby / noise pollution (1)

(tidal power)

no marks are given for choosing the correct non-renewable energy source

- barrages / visual pollution (1)

- destroys the habitat of many living organisms (1)

(hydroelectricity)

no marks are given for choosing the correct non-renewable energy source

- damming / visual pollution (1)
- very large areas of land (1) flooding (1)

4

[6]

Q2.

(a) The use of an incorrect equation scores zero. The substitution mark is only credited when the correct equation has been used. The second mark is for the rearrangement, the third mark is for the correct answer. An answer of 0.0025 (m) would score 2 marks for not converting the energy into J. An incorrect conversion (power of ten error) would score 2 marks. 77% of students scored 3 marks. 9% of students scored 2 marks. 12% of students scored zero marks.

(b) Both points seemed to score equally well, but very few students referred to the work done by the resistive forces. 19% of students scored 2 marks. 21% scored 1 mark. Sound energy was ignored.

(c) Since the question concerned the transfer of gravitational potential energy to kinetic energy, a difference in mass would have no effect on the final kinetic energy, $mgh = \frac{1}{2} mv^2$ therefore, $v^2 = 2gh$. So a student's explanation identifying a different mass leading to a different speed from energy considerations only, would score zero. An explanation that involved terminal velocity could score 2 marks. If a student stated that a higher mass would affect the speed, this would score 1 mark. Only 2% of students scored 2 marks. 39% of students scored 1 mark. 59% of students scored zero marks.

Q4.

(a) (i) This was the best answered numerical question with many candidates able to transform the equation and substitute values to arrive at the correct answer.

(ii) This proved difficult for the majority of candidates, even with the correct answer to the previous part.

(b) Very few correct answers were seen, the majority being too imprecise and not giving a reason for the variation in sunlight.

Q5.

(a) (i) Less than half of the candidates answered this correctly. Many were unable to transpose the equation correctly.

(ii) Only about a third of candidates were able to draw something resembling a Sankey diagram and label it appropriately. Whilst the majority of candidates obviously had some idea that a Sankey diagram had some arrows going in different directions, some drew a picture of solar cells, and over a tenth did not attempt the question.

(b) This question was correctly answered by the majority of candidates. However, many seem to have a dim view of company scientists, indicating that they would lie or falsify results.

(c) Nearly all candidates gained at least one mark, with more than half scoring both. The most common answers were that the Sun is a renewable resource and that solar cells do not produce pollutant gases.

Q6.

(a) This question was well answered with half the students scoring 1 mark and a third of students scoring 2 marks. The most common correct answers referred to the cost per kWh and the economic benefits, 'France can sell their excess electricity to other countries' type of statement. Insufficient responses included 'it's cheap', which wasn't comparative; or references to no CO2 released, as the renewables mentioned don't release CO2 either. Reliability was another commonly seen response, which was creditworthy.

(b) Just under a third of students scored 2 marks for this question. Answers that were insufficient were 'dangerous', or 'radiation may leak'. Naming nuclear accidents was insufficient for a mark, the idea of widespread or major implications was necessary too. Commissioning or decommissioning time was insufficient as the question was about generating electricity, so while the cost was an issue, time was too vague. A number of students thought that 'nuclear is a fossil fuel so contributes greatly to global warming', which is clearly incorrect.

(c) Three fifths of students scored 2 marks for this question. Some students incorrectly multiplied their answer by 100(%) and got an answer of 48, which scored 1 mark, or multiplied the power in W by 0.15 and got an answer of 480, which also scored 1 mark.

(d) Three fifths of students scored 1 mark for the idea that higher cost meant higher efficiency solar panel, quite a lot of students also scored 1 mark for the idea that if cheaper, more would be bought. Many students, however, incorrectly thought that if you purchased a larger number of solar panel C, the overall efficiency would increase. These students are likely to have scored a maximum of 1 mark for the idea that more could be bought, depending how they worded their answer. Only a tenth of students scored 2 marks for a well-reasoned answer e.g. The more efficient solar panels cost more, but you could buy more solar panel C for £40, that would generate more electricity than 1 solar panel A.

Q8.

(a) Most candidates could describe the unsuitability of using fossil fuels. Those who chose nuclear power often did not gain credit because they did not refer specifically to the problems of nuclear radiation and disposal of radioactive waste.

(b) Many candidates scored well on this part. Most could describe the problems involved in the use of renewable energy sources, although most confined their answers to visual pollution.